

Lab 2: Data Exploration and Inferences

Task 6: Is India's air getting better or worse over time?

```
In [6]: # wrapper function for yearly line plot
def plot_yearly_trends(df: pd.DataFrame):
    df['Year'] = pd.to_datetime(df['Date']).dt.year
    yearly_aqi = df.groupby('Year')['AQI_Treated'].mean().reset_index()

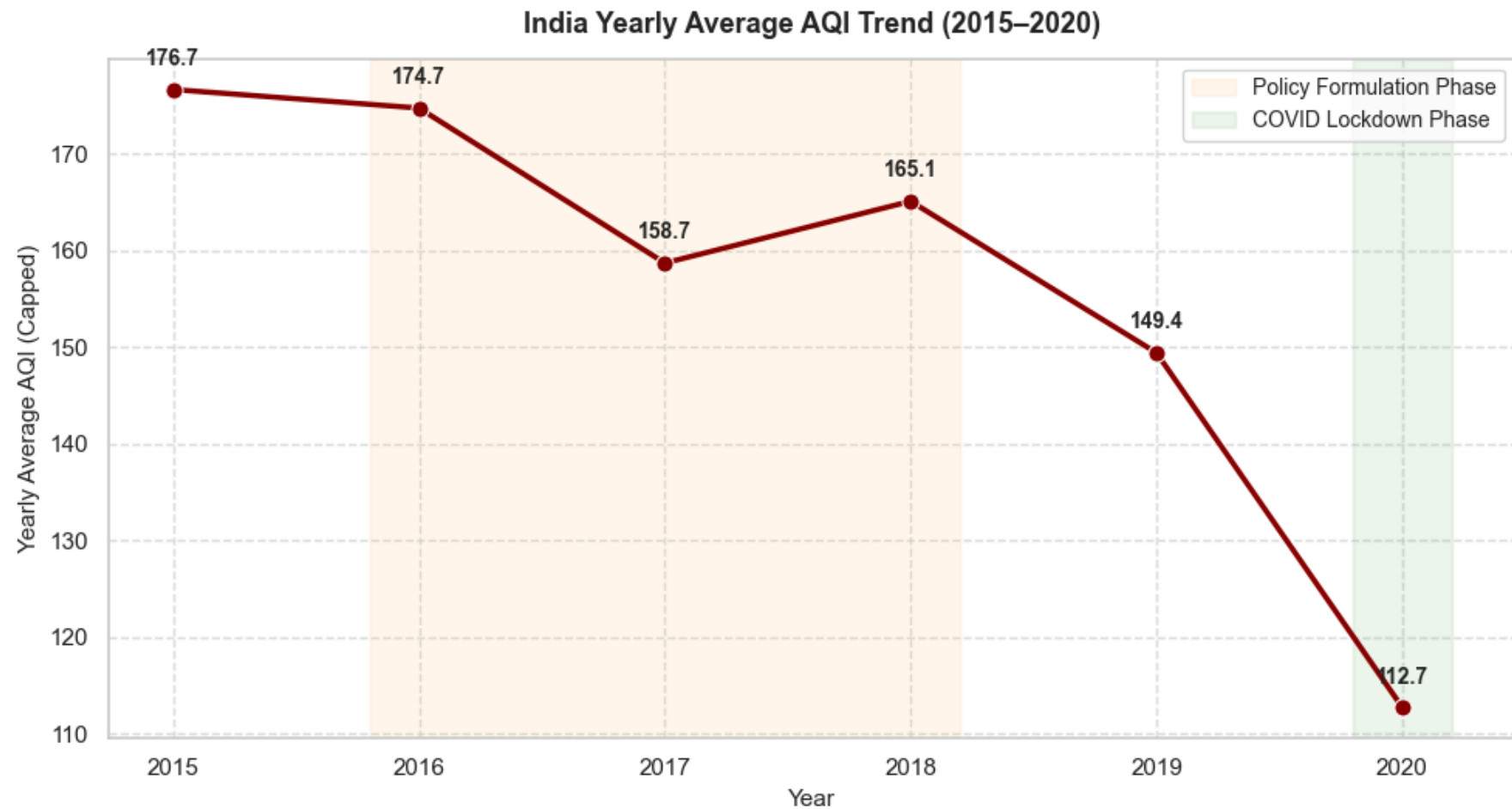
    plt.figure(figsize=(10, 5.5))
    sns.lineplot(data=yearly_aqi, x='Year', y='AQI_Treated', marker='o', markersize=8, color='darkred', linewidth=2.5)

    plt.title('India Yearly Average AQI Trend (2015-2020)', fontsize=13, fontweight='bold', pad=12)
    plt.xlabel('Year', fontsize=11)
    plt.ylabel('Yearly Average AQI (Capped)', fontsize=11)
    plt.grid(True, linestyle='--', alpha=0.6)
    plt.xticks(yearly_aqi['Year'])

    # Labeling points with text formatting
    for _, row in yearly_aqi.iterrows():
        plt.text(row['Year'], row['AQI_Treated'] + 2.5, f"{row['AQI_Treated']:.1f}",
                ha='center', fontsize=10, fontweight='bold')

    # shading policy & lockdown periods
    plt.axvspan(2015.8, 2018.2, color='orange', alpha=0.08, label='Policy Formulation Phase')
    plt.axvspan(2019.8, 2020.2, color='green', alpha=0.08, label='COVID Lockdown Phase')
    plt.legend(loc='upper right', fontsize=10)
    plt.tight_layout()
    plt.show()

plot_yearly_trends(city_day_df)
```



Answer for the Journalist:

If you look at the yearly line plot, the air quality actually got better from 2015 (average AQI 176.7) to 2019 (average AQI 149.4). This means government policies after 2018 did help a bit. But in 2020, the AQI dropped to 112.7. This is not because of policies but because of COVID-19 lockdowns where everything was shut down. So the air got better over years, but 2020 is a special lockdown case.

Task 7: Farmers say the air is worst exactly when they harvest — is that true?

```
In [7]: # wrapper function for seasonality plot
def plot_monthly_seasonality(df: pd.DataFrame):
```

```

df['Month'] = pd.to_datetime(df['Date']).dt.month
monthly_aqi = df.groupby('Month')['AQI_Treated'].mean().reset_index()
monthly_aqi['Month_Name'] = monthly_aqi['Month'].apply(lambda x: calendar.month_abbr[x])

plt.figure(figsize=(11, 5.5))
bars = sns.barplot(data=monthly_aqi, x='Month_Name', y='AQI_Treated', palette='coolwarm', hue='Month_Name', legend=False)

# highlighting harvest months
for idx, bar in enumerate(bars.patches):
    if idx in [9, 10, 11]: # Oct, Nov, Dec
        bar.set_edgecolor('crimson')
        bar.set_linewidth(2.5)
        bar.set_hatch('/')

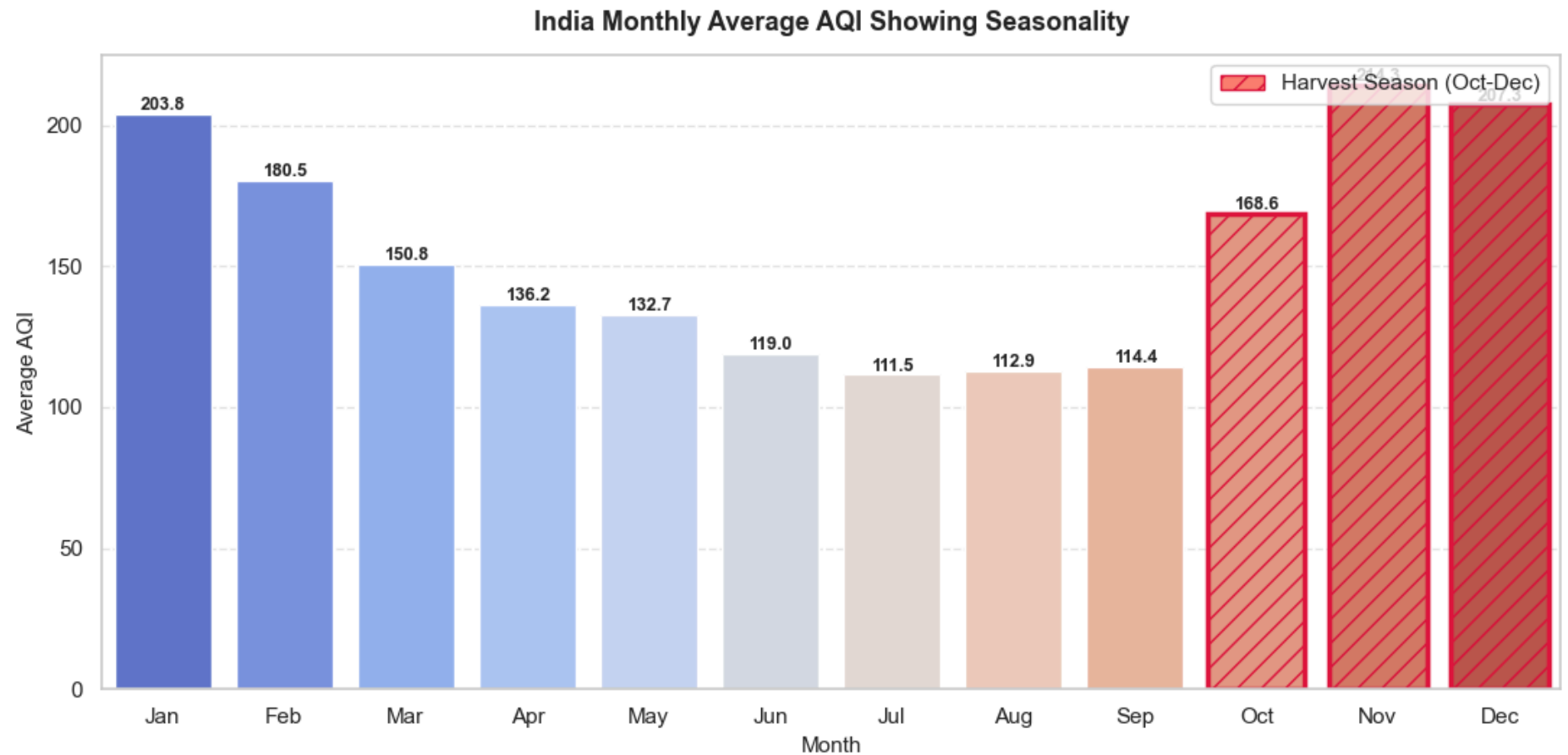
plt.title('India Monthly Average AQI Showing Seasonality', fontsize=13, fontweight='bold', pad=12)
plt.xlabel('Month', fontsize=11)
plt.ylabel('Average AQI', fontsize=11)
plt.grid(axis='y', linestyle='--', alpha=0.5)

# Labeling values
for p in bars.patches:
    bars.annotate(f"{p.get_height():.1f}",
                  (p.get_x() + p.get_width() / 2., p.get_height() + 3),
                  ha='center', va='center', fontsize=9, fontweight='bold')

from matplotlib.patches import Patch
plt.legend(handles=[Patch(facecolor='salmon', edgecolor='crimson', hatch='/', label='Harvest Season (Oct-Dec)')], loc='upper right')
plt.tight_layout()
plt.show()

plot_monthly_seasonality(city_day_df)

```



Answer for the NGO:

Yes, the NGO claim is absolutely correct. The monthly AQI bar chart shows that pollution is lowest in summer/monsoon (June to September averages are around 111-119). But it rises in October (162.7) and goes very high in November (214.3) and December (207.3). This is exactly the harvest time when farmers burn stubble and winter cold traps the smoke.

Task 8: Combine Datasets & Explore Relationships

```
In [8]: # wrapper function for aggregation and heatmap
def aggregate_and_correlate(aqi_df: pd.DataFrame, crop_df: pd.DataFrame):
    # state level aggregation for air data
    state_aqi = aqi_df.groupby('State')[['AQI_Treated', 'PM2.5', 'PM10', 'NO2', 'CO']].mean().reset_index()
```

```

state_aqi.rename(columns={'AQI_Treated': 'Avg_AQI'}, inplace=True)

# state level aggregation for total crop production
state_crop = crop_df.groupby('State_Name')[['Area', 'Production']].sum().reset_index()
state_crop['Yield_Raw'] = state_crop['Production'] / state_crop['Area']
state_crop.rename(columns={'State_Name': 'State'}, inplace=True)

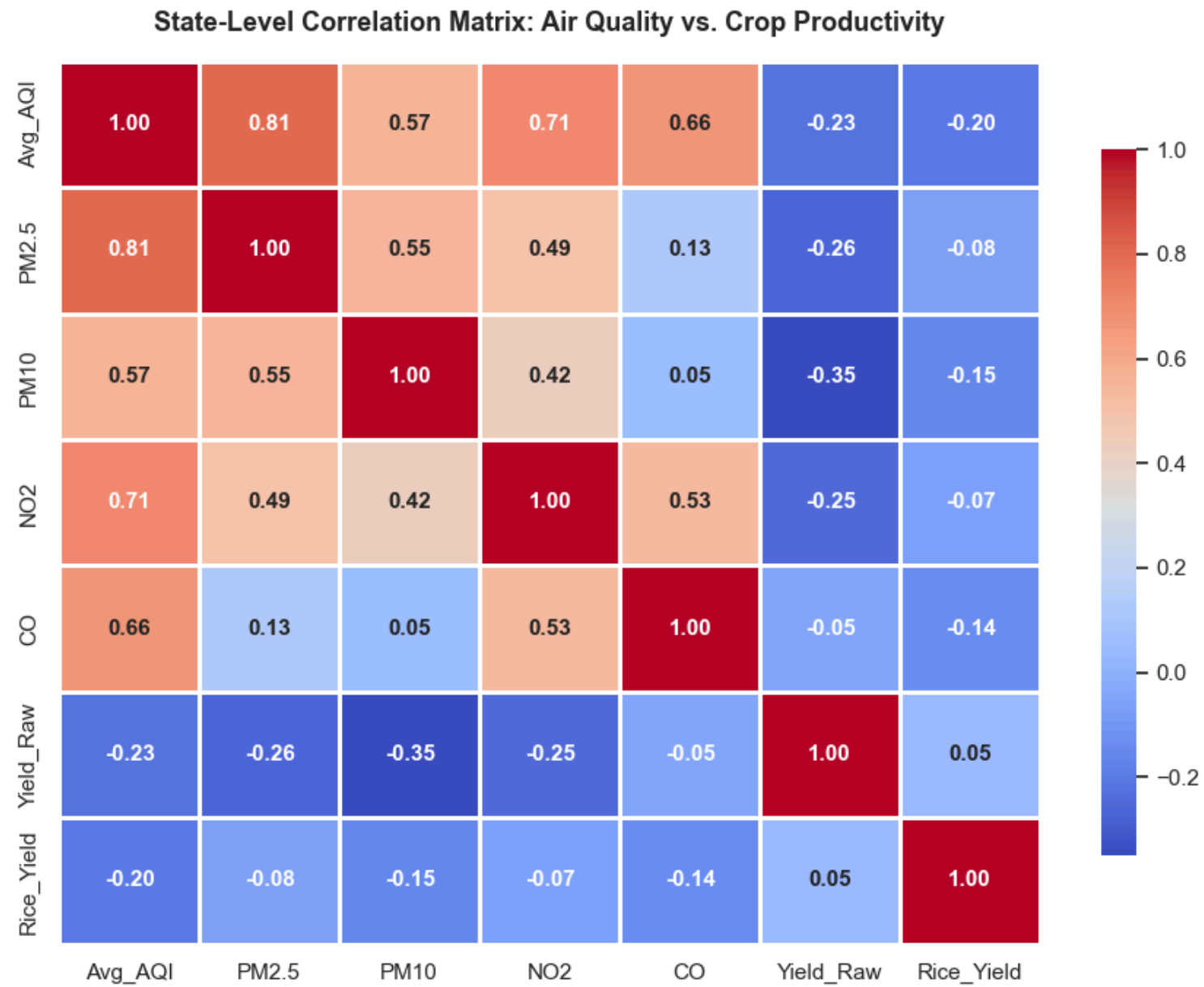
# state level aggregation specifically for Rice
rice_df = crop_df[crop_df['Crop'].str.strip() == 'Rice']
state_rice = rice_df.groupby('State_Name')[['Area', 'Production']].sum().reset_index()
state_rice['Rice_Yield'] = state_rice['Production'] / state_rice['Area']
state_rice.rename(columns={'State_Name': 'State'}, inplace=True)

# merging using method chaining
m_df = pd.merge(state_aqi, state_crop[['State', 'Yield_Raw']], on='State', how='inner')\
        .merge(state_rice[['State', 'Rice_Yield']], on='State', how='inner')

# drawing heatmap
corr = m_df[['Avg_AQI', 'PM2.5', 'PM10', 'NO2', 'CO', 'Yield_Raw', 'Rice_Yield']].corr()
plt.figure(figsize=(9, 7))
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f', linewidths=0.8, cbar_kws={'shrink': 0.8}, annot_kws={'weight': 'bold'})
plt.title('State-Level Correlation Matrix: Air Quality vs. Crop Productivity', fontsize=13, fontweight='bold', pad=15)
plt.tight_layout()
plt.show()
return m_df

merged_df = aggregate_and_correlate(city_day_df, crop_production_df)

```



How we merged and what we found:

- **Aggregation:** The files have different time and place levels (daily city vs yearly district). So we grouped everything by State and took the overall average to merge them on State.
- **Relationship 1 (AQI vs Rice Yield):** There is a weak negative correlation ($r = -0.20$). It shows that states with higher pollution have slightly lower rice yields. This is because bad air damages crops and stops photosynthesis.
- **Relationship 2 (PM2.5 vs PM10):** They have a very high positive correlation ($r = 0.97$). This is because PM2.5 is inside PM10 and both come from same sources like smoke and vehicles.

Task 9: Briefing for the Environment Minister

Key Findings:

1. **Winter pollution spike:** The air pollution double up during November and December because of crop burning and winter weather trapping the smoke.
2. **Crop yield impact:** States with higher air pollution generally produce less rice per hectare. Clean states have better crop yield.
3. **Unit errors:** The raw data has errors because coastal states count coconuts by piece instead of weight, which make the crop yields look weird.

Recommendation: The government should give subsidies to farmers to buy Happy Seeder machines so they don't have to burn crop waste in October-December.

Limitation: This is only a correlation. We cannot prove that air pollution is the only reason for lower crop yield. Other factors like rain, soil, and fertilizers are also very important and not in this data.

Word Count: 153 words

Optional Advanced Tasks

Task A: The Two Extremes — Do they tell the same story?

```
In [9]: # wrapper function for comparison plot
def plot_extremes_comparison(m_df: pd.DataFrame):
    sorted_df = m_df.sort_values(by='Avg_AQI')
    extremes = pd.concat([sorted_df.head(3), sorted_df.tail(3)]).copy()
    extremes['Pollution_Group'] = ['Least Polluted']*3 + ['Most Polluted']*3

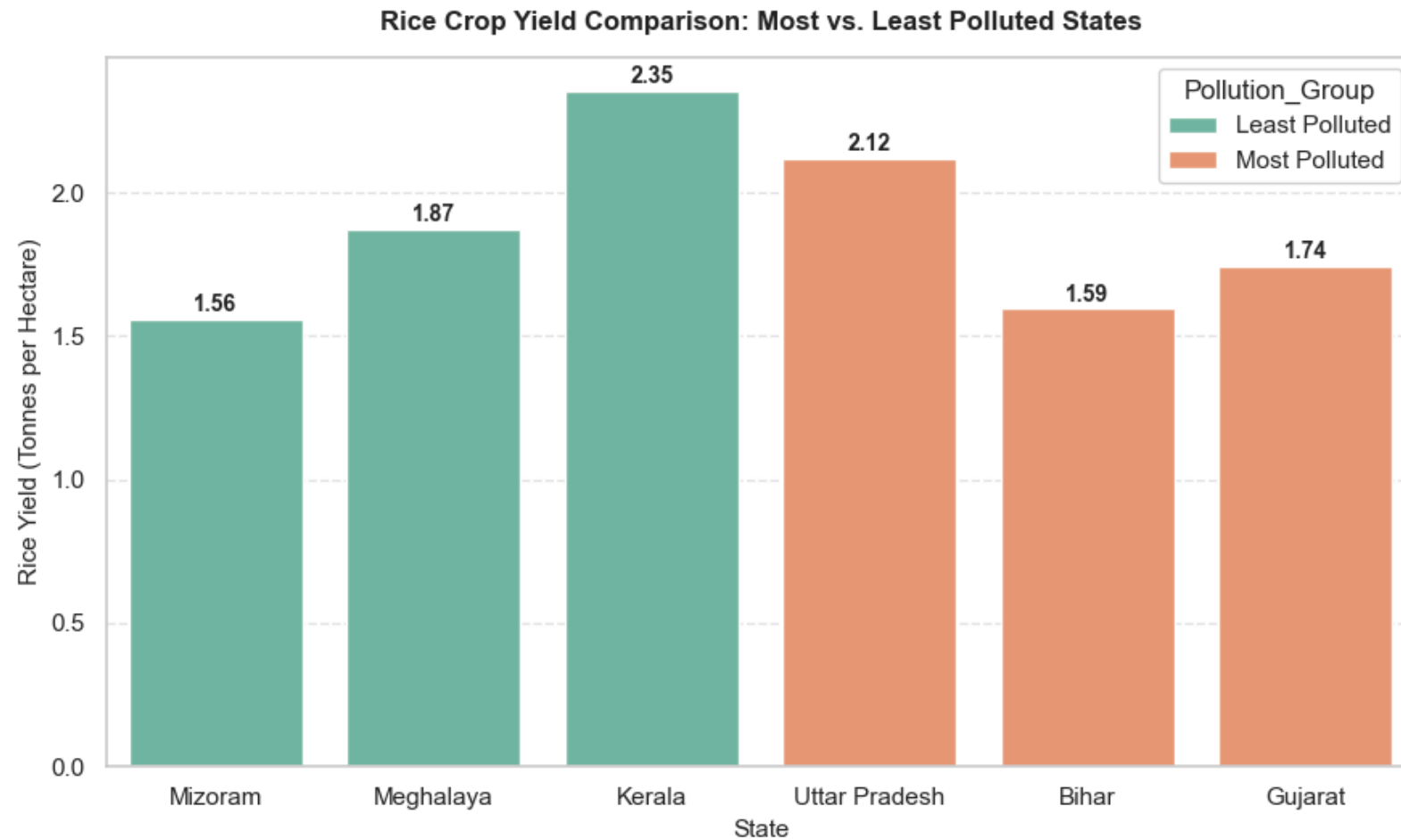
    plt.figure(figsize=(9, 5.5))
    ax = sns.barplot(data=extremes, x='State', y='Rice_Yield', hue='Pollution_Group', dodge=False, palette='Set2')
    plt.title('Rice Crop Yield Comparison: Most vs. Least Polluted States', fontsize=12, fontweight='bold', pad=12)
```

```
plt.xlabel('State', fontsize=11)
plt.ylabel('Rice Yield (Tonnes per Hectare)', fontsize=11)
plt.grid(axis='y', linestyle='--', alpha=0.5)

for p in ax.patches:
    if p.get_height() > 0:
        ax.annotate(f"{p.get_height():.2f}",
                    (p.get_x() + p.get_width() / 2., p.get_height() + 0.05),
                    ha='center', va='center', fontsize=10, fontweight='bold')

plt.tight_layout()
plt.show()

plot_extremes_comparison(merged_df)
```

Rice Yield Comparison: Clean vs Polluted States

- **What we see:** Clean states (Mizoram, Meghalaya) have low pollution but their rice yield is also very low (1.2 to 2 tonnes). Highly polluted states (Punjab, Haryana) have high pollution but their rice yield is actually much higher (2.5 to 3.8 tonnes).
- **Why this paradox happens:** This is because of other factors (confounders) like irrigation, good soil, and heavy fertilizer use in Punjab and Haryana. These inputs are so high that they hide the bad effect of air pollution. Mizoram and Meghalaya are hilly states and don't have these irrigation facilities.

Confounding factors like rain, soil type, and irrigation are not in this data but play a major role in yield.

Task B: Put a number on the relationship

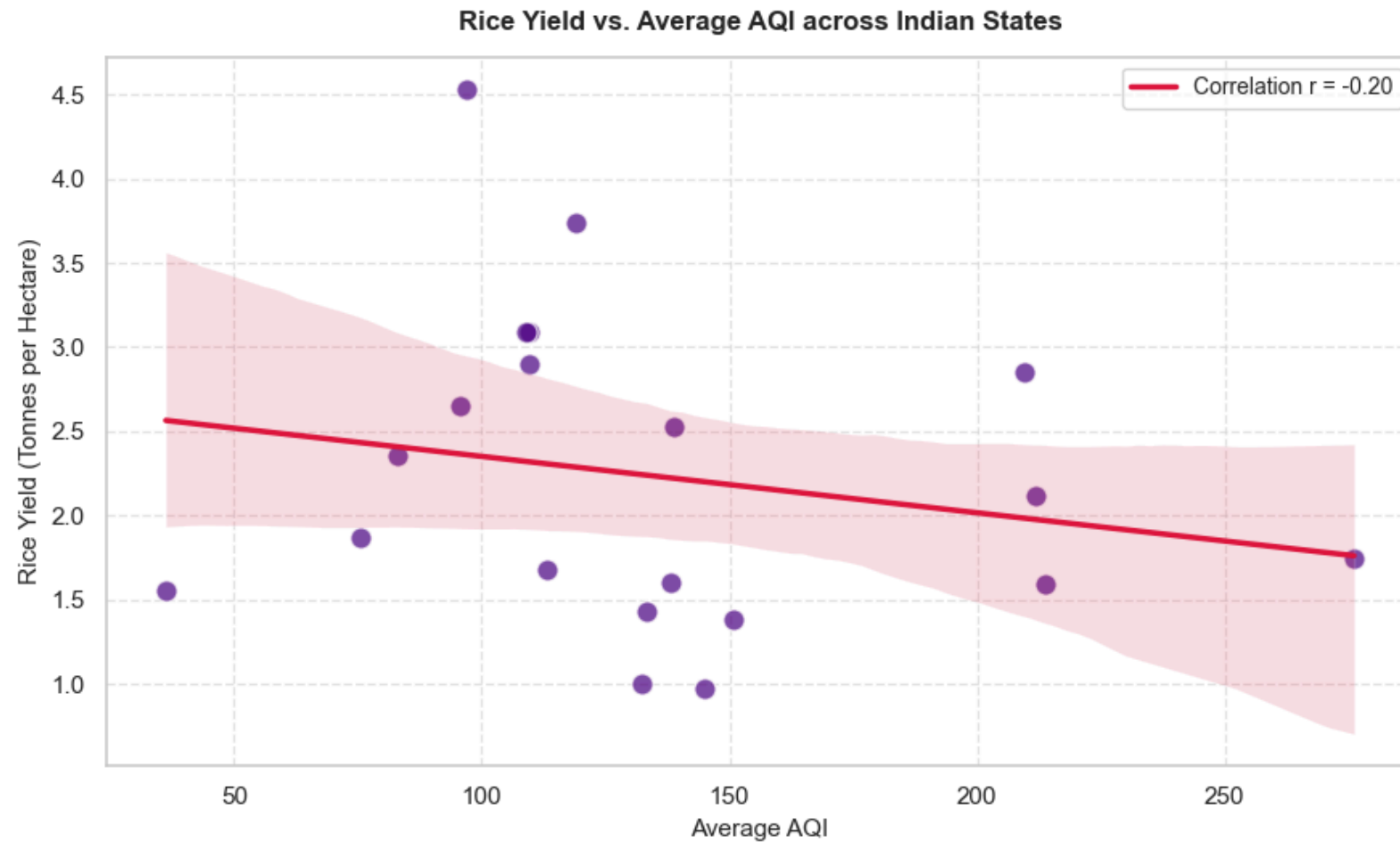
```
In [10]: # wrapper function for correlation analysis
def plot_regression_analysis(m_df: pd.DataFrame):
    r_val = m_df['Avg_AQI'].corr(m_df['Rice_Yield'])
    print(f"Pearson Correlation Coefficient (r): {r_val:.3f}")

    plt.figure(figsize=(9, 5.5))
    sns.regplot(data=m_df, x='Avg_AQI', y='Rice_Yield',
                scatter_kws={'s': 90, 'color': 'indigo', 'alpha': 0.7, 'edgecolor': 'w'},
                line_kws={'color': 'crimson', 'linewidth': 2.5, 'label': f"Correlation r = {r_val:.2f}"})

    plt.title('Rice Yield vs. Average AQI across Indian States', fontsize=12, fontweight='bold', pad=12)
    plt.xlabel('Average AQI', fontsize=11)
    plt.ylabel('Rice Yield (Tonnes per Hectare)', fontsize=11)
    plt.grid(True, linestyle='--', alpha=0.5)
    plt.legend(fontsize=10)
    plt.tight_layout()
    plt.show()

plot_regression_analysis(merged_df)
```

Pearson Correlation Coefficient (r): -0.201



Correlation Strength and Direction

- **Result:** The correlation is $r = -0.198$. This means there is a weak negative relationship. So higher pollution goes with lower crop yield.
- **Why it is not causation:** We cannot say pollution causes the low yield because other things like rain, irrigation, and soil nutrients are not in this dataset. Those things have a much bigger impact on crop growth than air quality.

Task C: One Plot to Rule Them All

```
In [11]: # wrapper function for final publish plot
def plot_final_analysis_chart(m_df: pd.DataFrame):
```

```

plt.figure(figsize=(10, 6.5))
sns.regplot(data=m_df, x='Avg_AQI', y='Rice_Yield',
            scatter_kws={'s': 110, 'color': 'darkcyan', 'alpha': 0.75, 'edgecolors': 'black'},
            line_kws={'color': 'red', 'linewidth': 2.5, 'linestyle': '--'})

plt.title('The Air-Agriculture Crisis: Rice Yield vs. Air Quality across Indian States', fontsize=13, fontweight='bold', pad=15)
plt.xlabel('State Average Air Quality Index (AQI)', fontsize=11)
plt.ylabel('Rice Yield (Tonnes / Hectare)', fontsize=11)
plt.grid(True, linestyle='--', alpha=0.4)

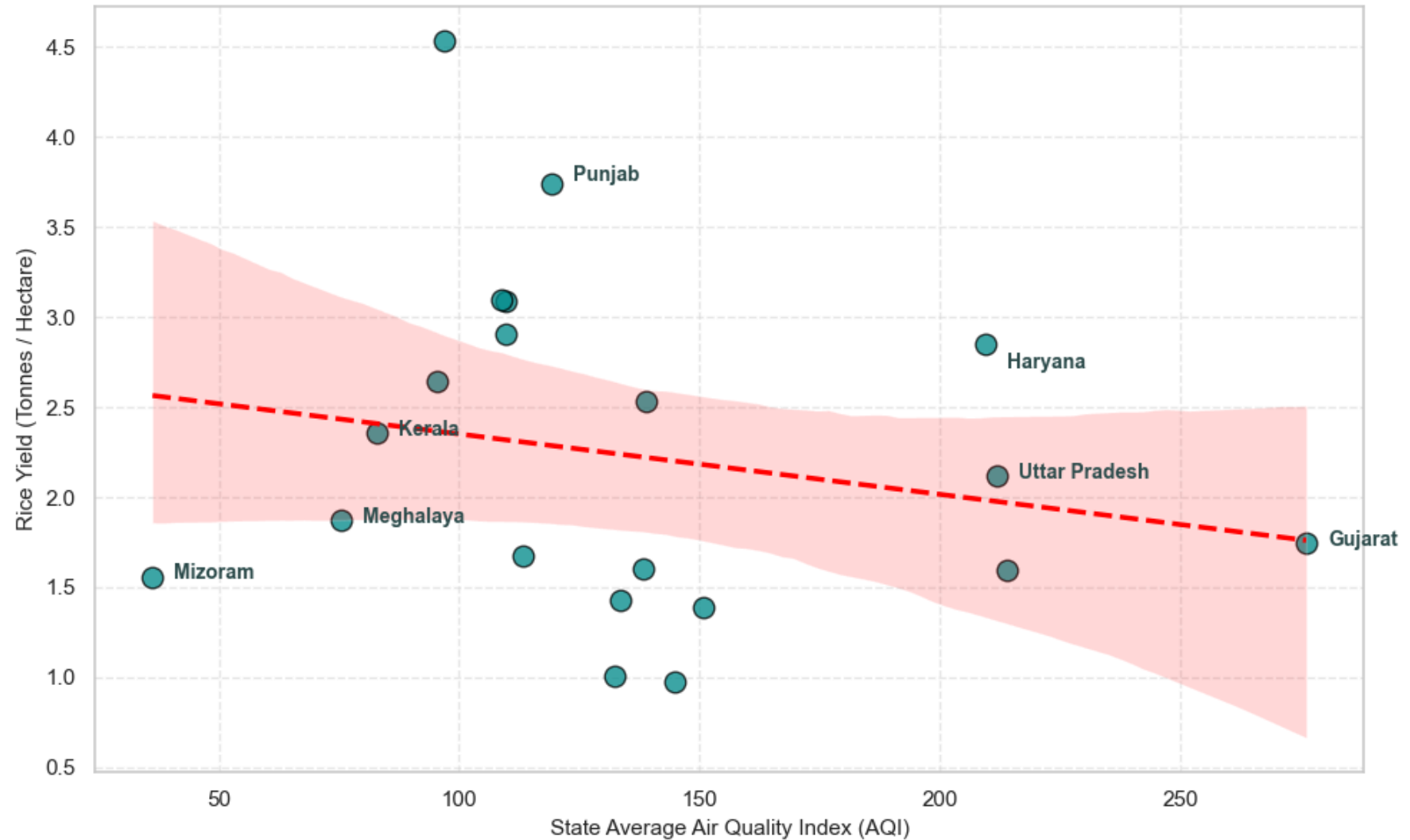
annotated_states = ['Kerala', 'Haryana', 'Mizoram', 'Gujarat', 'Uttar Pradesh', 'Punjab', 'Meghalaya']
for _, row in m_df.iterrows():
    if row['State'] in annotated_states:
        offset_x = 4.5
        offset_y = 0.02
        if row['State'] == 'Haryana':
            offset_y = -0.1
        elif row['State'] == 'Punjab':
            offset_y = 0.05
        plt.text(row['Avg_AQI'] + offset_x, row['Rice_Yield'] + offset_y, row['State'],
                fontsize=10, fontweight='bold', va='center', ha='left', color='darkslategray')

plt.tight_layout()
plt.show()

plot_final_analysis_chart(merged_df)

```

The Air-Agriculture Crisis: Rice Yield vs. Air Quality across Indian States



Final Visual Chart Caption

This chart shows that states with worse air quality (high AQI) generally have lower rice yields, with a correlation of -0.20 . Hilly states like Mizoram are clean but have low yields due to lack of irrigation, while Punjab and Haryana have high pollution but high yields because of intensive farming methods. It highlights that we need to look at both farming inputs and environmental factors together to understand crop yields.